Voltage Protector Device - Project Report

FSMB Recruitment 2026, Phase 1 - Embedded System Engineer

Overview

A voltage protector on an STM32F103 (Blue Pill) that disconnects a load within one 100 ms sample of an over-voltage, under-voltage or over-current fault, and lets the user change the limits from the device or from a computer. It was built and tested in the Wokwi simulator; potentiometers stand in for the sensors, so nothing touches mains.

- **Protection:** defaults 250 V over, 170 V under, 10 A over-current; any fault drops the relay.
- **Recovery:** automatic once the reading is back in range plus a 3 s cooldown, or early with the RESET button.
- **Local settings:** SEL / UP / DOWN change OV, UV and OC (5 V or 0.5 A steps); the LCD shows live values and the setpoint being edited.
- **Computer settings:** UART commands SET_OV, SET_UV, SET_OC, STATUS, RESET, HELP.
- **Continuous logging:** status line goes out over serial every second; the heartbeat LED blinks at 1 s normally, 200 ms in fault

Components and why

Part	Pin	Role	Why
STM32F103 Blue Pill	-	Runs protection logic at 64 MHz (HSI/2 x 16)	12-bit ADC, UART and enough GPIO on one cheap board
Potentiometer 1	PA0 (ADC0)	Voltage sensor stand-in, 0-330 V	Safe low-voltage emulation of the grid
Potentiometer 2	PA1 (ADC1)	Current sensor stand-in, 0-20 A	Lets any fault be injected by hand
Relay (LED + 220 ohm in sim)	PB0	High = load connected, low = disconnected	Cuts the load on any fault; off until the FSM reconnects
RESET button	PB12	Skips the wait after a fault	Operator override, active-low
SEL / UP / DOWN buttons	PB13/14/15	Choose a setting and change its value	Settings without a computer
16x2 LCD, I2C backpack	PB6 SCL, PB7 SDA	Shows readings, state, setting being edited	I2C needs 2 wires instead of 6
Serial link (USART1)	PA9 TX, PA10 RX	Logging and computer commands	Works with any terminal or USB-serial adapter
Heartbeat LED	PC13	Shows the main loop is alive	Quick health check

Protocols

Interface	Protocol	Details
Serial monitor / commands	UART 115200, 8N1	ASCII lines ending CR/LF. TX status line every 1 s. RX interrupt into a ring buffer, parsed in main loop.
LCD	I2C, PCF8574 at 0x27	Bit-banged on PB6/PB7 with GPIO, HD44780 in 4-bit mode. Display is not time-critical.
Voltage / current sensing	ADC, 12-bit	Polled single conversions, 55.5-cycle sample time, scaled by 330/4095 V and 20/4095 A.
Buttons / relay	GPIO	Buttons polled, active-low with pull-ups. Relay is a push-pull output.

How it works and results

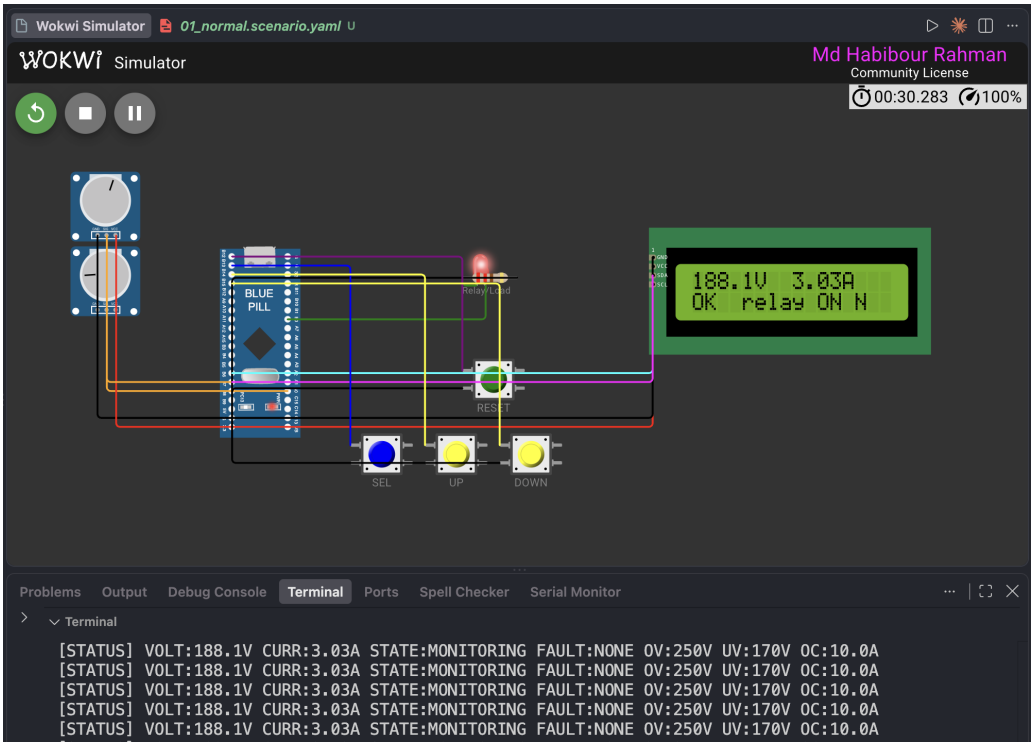
The main loop is cooperative and polled. Every 100 ms it reads both ADC channels and calls a state machine (MONITORING, DISCONNECTED, COOLDOWN) that drives the relay; the UI and UART modules run on every pass. Code is split into sensors, fsm, relay, settings, ui and uart_comm, and every new limit is range-checked (OV above UV and at most 330 V; OC above 0 and at most 20 A).

Ten scenarios were run with wokwi-cli; eight passed and the two UART scenarios could not be verified in the simulator.

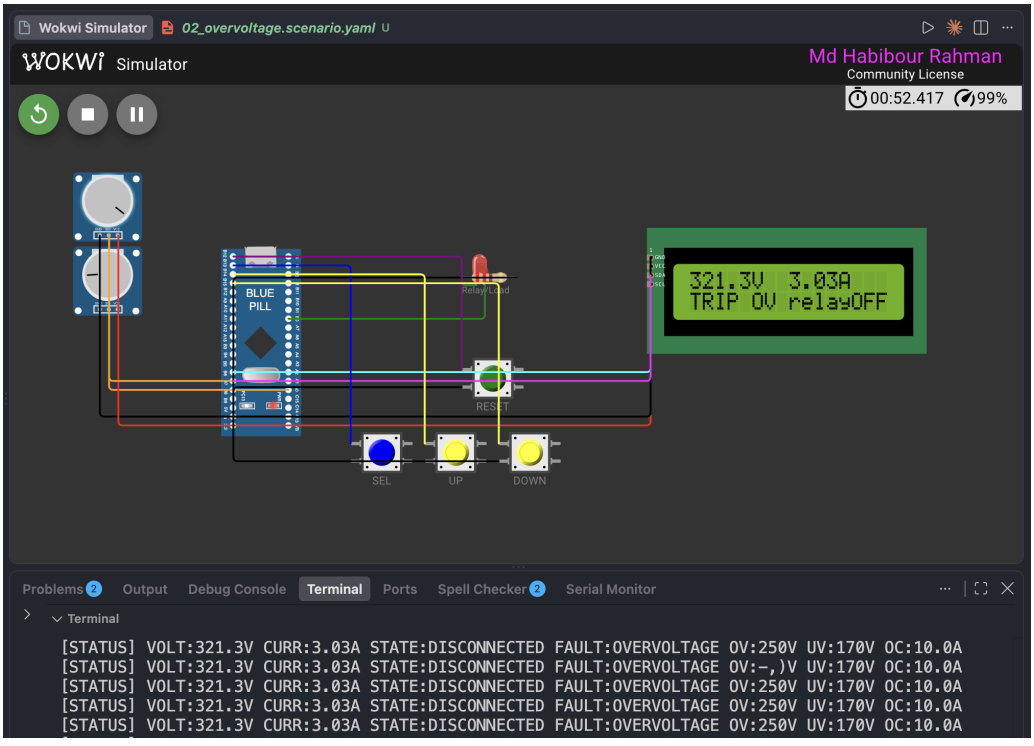
ID	Scenario	Result
01	Normal operation, 231 V / 5 A	PASS. MONITORING, relay on (LED IS ON)
02	Over-voltage, 280 V	PASS. DISCONNECTED / OVERVOLTAGE, relay off (LED IS OFF)
03	Under-voltage, 132 V	PASS. DISCONNECTED / UNDERVOLTAGE, relay off. (LED IS OFF)
04	Over-current, 15 A	PASS. DISCONNECTED / OVERCURRENT, relay off. (LED IS OFF)
05	Cooldown after fault clears	PASS. COOLDOWN (relay off), then MONITORING (relay on). (LED IS ON)
06	Reset button (PB12) while faulted	PASS. Goes to COOLDOWN; a persistent fault trips again.
07	UART commands	NOT VERIFIED. Wokwi did not deliver serial input to USART1 RX.
08	UART setting takes effect	NOT VERIFIED. Same cause as 07.
09	Response time, over-voltage at 5010 ms	PASS. Relay off at about 5044 ms (34 ms), within one 100 ms sample.
10	On-device UI: SEL, UP, UP on OV	PASS. LCD shows SET OVERVOLTAGE 260V.

Appendix - Simulation screenshots

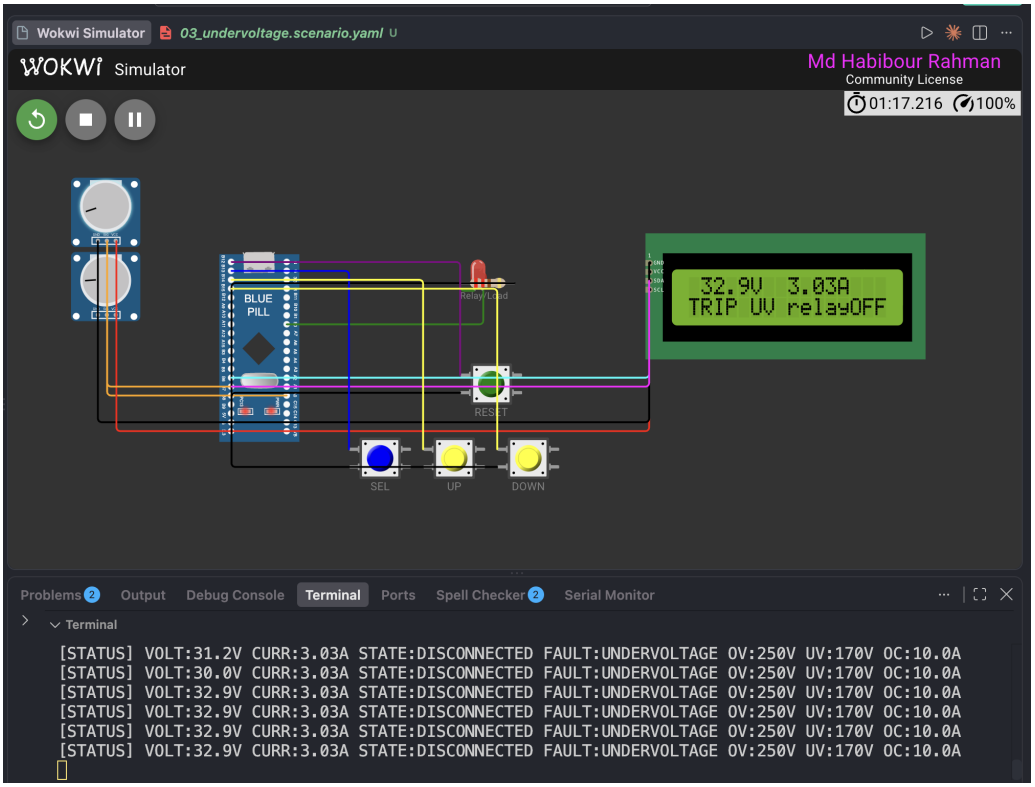
Wokwi simulation of the firmware, one capture per scenario. Potentiometer 1 = voltage (0-330 V), potentiometer 2 = current (0-20 A); the red LED on PB0 is the relay.



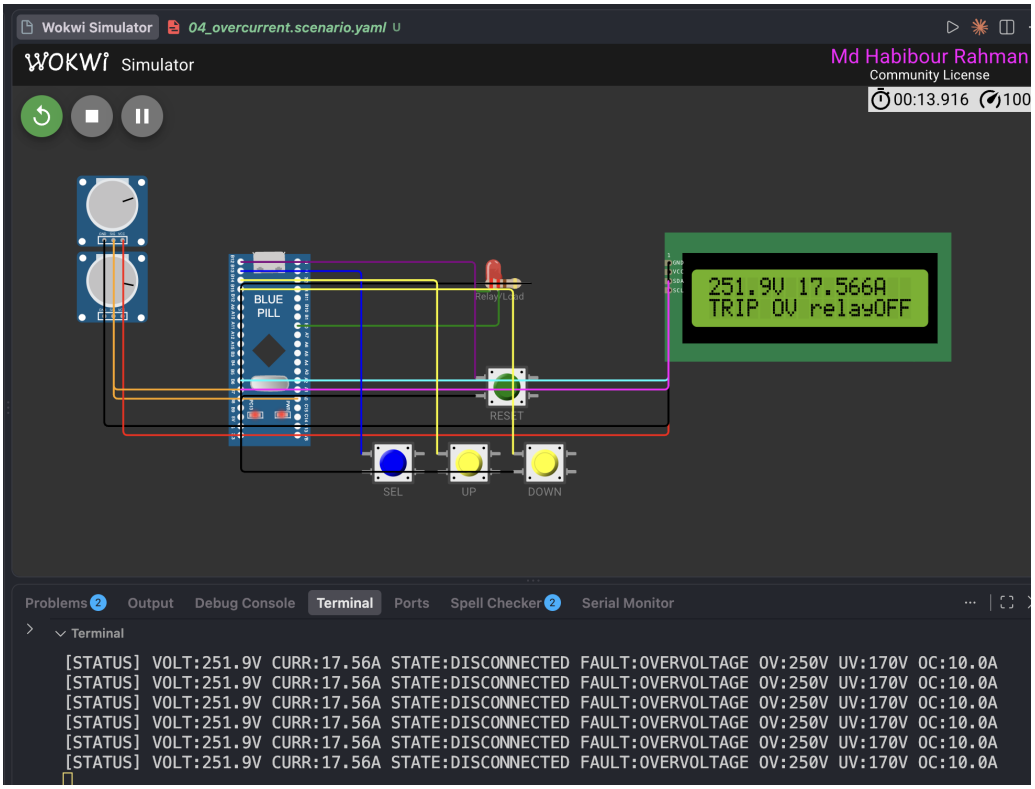
01 Normal operation. 188.1 V, 3.03 A inside the limits. State MONITORING, LCD *OK relay ON*. Relay LED on.



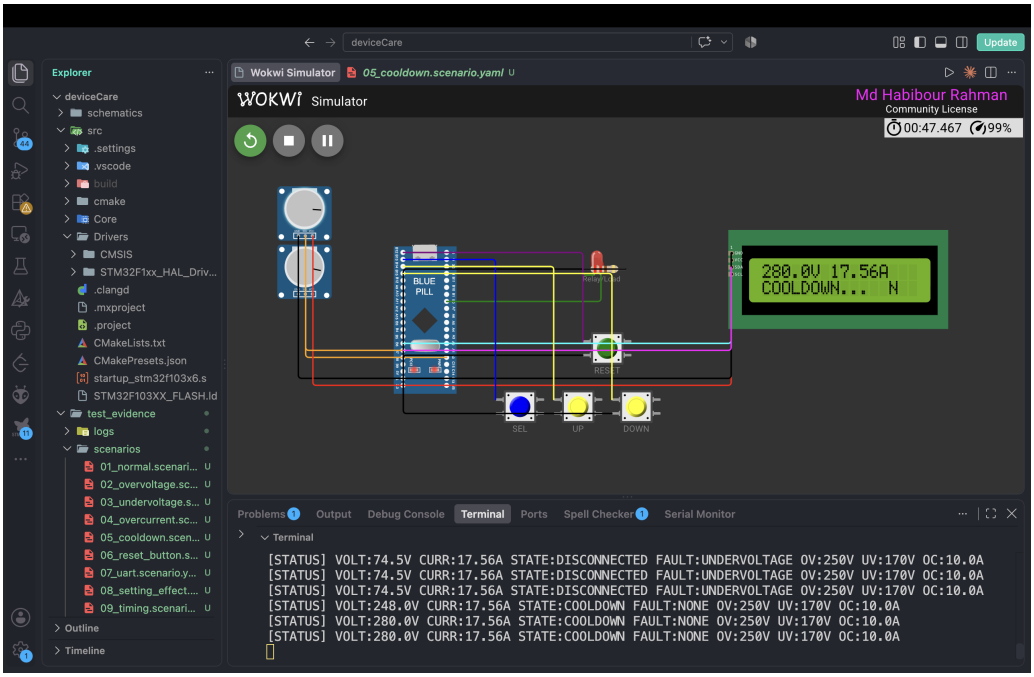
02 Over-voltage. 321.3 V is above the 250 V limit. State DISCONNECTED, LCD *TRIP OV relayOFF*. Relay LED off.



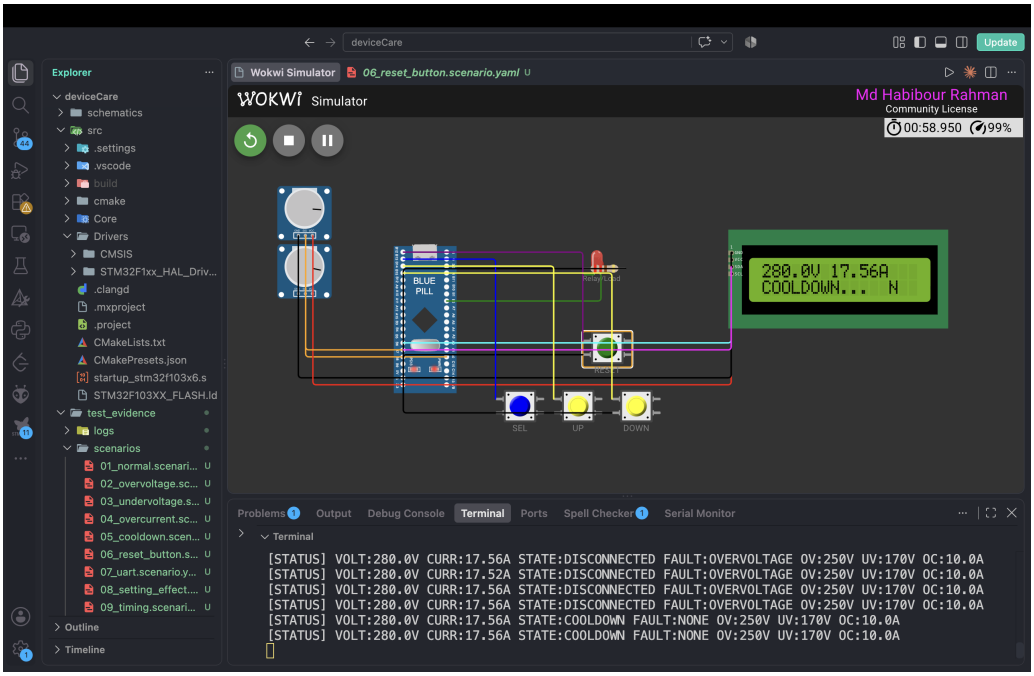
03 Under-voltage. 32.9 V is below the 170 V limit. State DISCONNECTED, LCD *TRIP UV relayOFF*. Relay LED off.



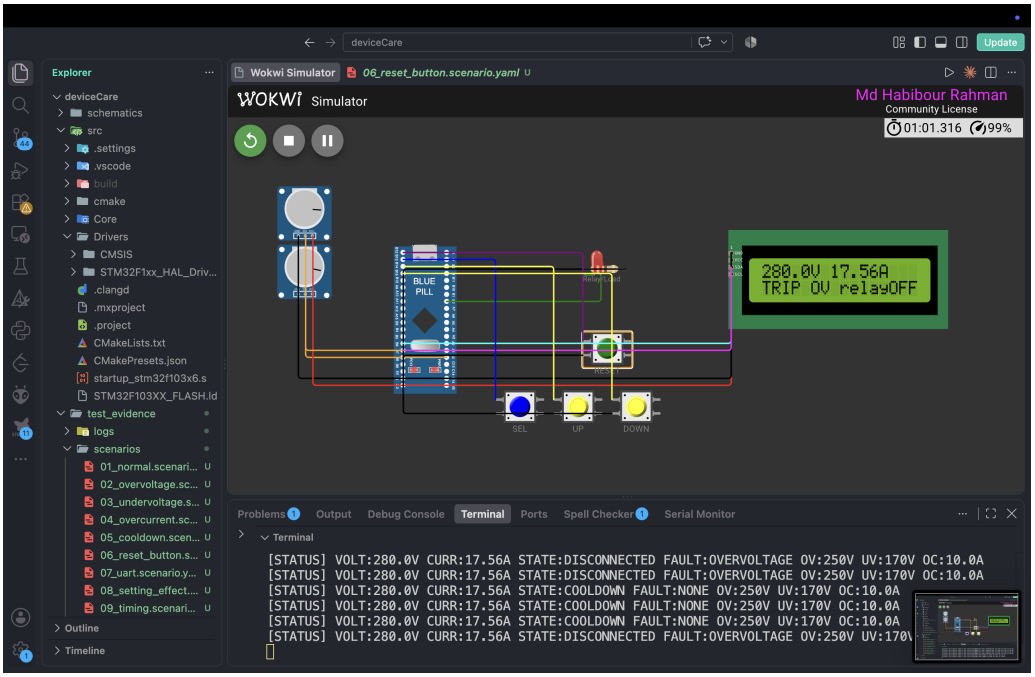
04 Over-current. 17.56 A is above the 10 A limit and the relay is off. Voltage was also 251.9 V, so the fault shown is OV (voltage is checked first); the over-current-only case is in logs/04_overcurrent.log. Relay LED off.



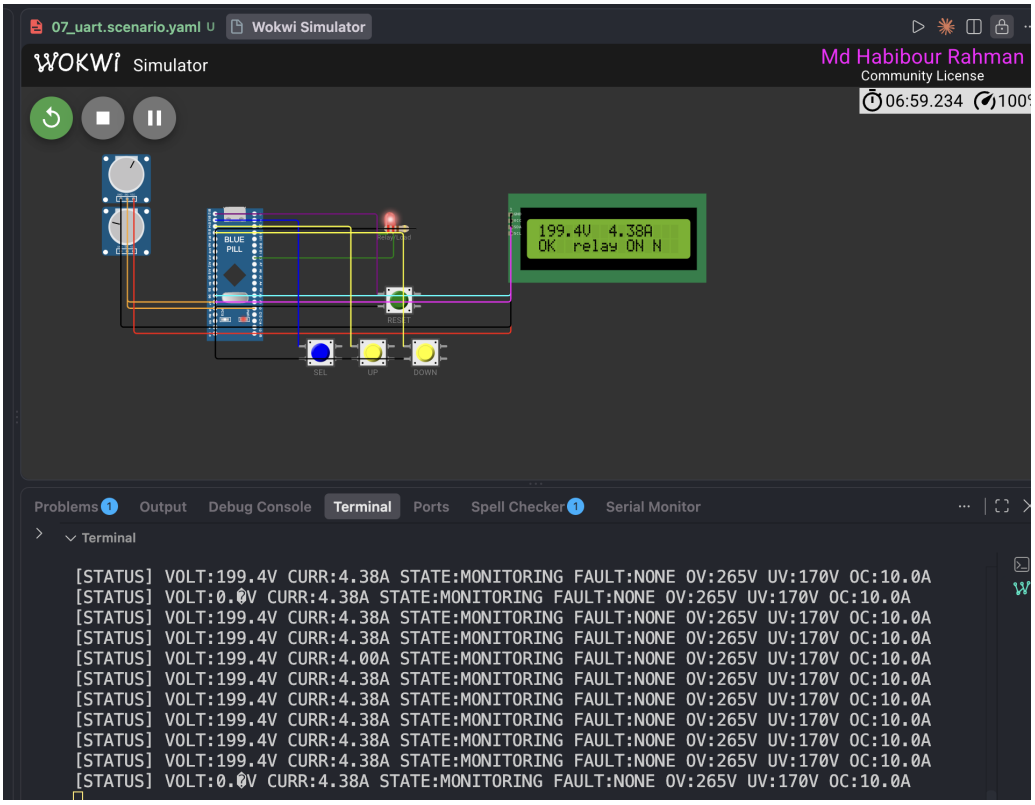
05 Cooldown. After the under-voltage clears the state is COOLDOWN, LCD *COOLDOWN...*. Relay LED stays off for 3 s, then reconnects.



06 Reset button, pressed. With an over-voltage fault active, RESET is pressed and the state goes to COOLDOWN. Relay LED off.



06 Reset button, outcome. 280 V is still applied, so after the cooldown the device trips again: DISCONNECTED / OVERVOLTAGE, LCD *TRIP OV relayOFF*.



07 Limit changed. The status line reports OV:265V instead of the 250 V default, with 199.4 V inside the window: MONITORING, relay on. UART input could not be verified in Wokwi, so this does not show how the limit was set.